

Solution Guide – Google Play Store App Analysis (EDA Project)

This guide contains all Basic, Intermediate, and Advanced questions along with the approach used to solve them.

Basic Level Questions

1. What is the distribution of app ratings?

Approach

- Analyze the `Rating` column.
- Plot a histogram with KDE curve.

Solution

```
sns.histplot(df_cleaned['Rating'], bins=30, kde=True)
```

Insight

Most apps have ratings between 4.0 and 4.5.

2. Which categories have the most apps?

Approach

- Group by `Category`.
- Count the number of apps.

Solution

```
category_counts = df_cleaned['Category'].value_counts()
```

Insight

Family, Game, and Tools categories contain the largest number of apps.

3. What is the distribution of app sizes?

Approach

- Use the cleaned Size column.
- Plot a histogram.

Solution

```
sns.histplot(df_cleaned['Size'], bins=30, kde=True)
```

Insight

Most apps are below 20 MB.

4. How many free vs paid apps are there?

Approach

- Count values in the Type column.

Solution

```
df_cleaned['Type'].value_counts()
```

Insight

Free apps dominate the marketplace.

5. What is the distribution of content ratings?

Approach

- Count each Content Rating category.

Solution

```
df_cleaned['Content Rating'].value_counts()
```

Insight

Everyone category contains most applications.

6. Which are the top 5 most installed apps?

Approach

- Sort by Installs.
- Select top 5.

Solution

```
top5_apps = df_cleaned[['App', 'Installs']].sort_values(  
    by='Installs',  
    ascending=False  
) .head(5)
```

Insight

Google Drive, Facebook, and Instagram dominate installs.

7. What percentage of apps have ratings above 4.0?

Approach

- Count ratings ≥ 4.0 .
- Divide by total records.

Solution

```
percentage = (  
    (df_cleaned['Rating'] >= 4.0).sum()  
    / len(df_cleaned)  
) * 100
```

Insight

Most applications maintain ratings above 4.0.

8. Compare average reviews between free and paid apps.

Approach

- Group by Type.
- Calculate average Reviews.

Solution

```
avg_reviews = df_cleaned.groupby('Type')['Reviews'].mean()
```

Insight

Free apps receive significantly more reviews.

9. Which categories have the largest average app size?

Approach

- Group by Category.
- Calculate mean Size.

Solution

```
avg_size_category = (  
    df_cleaned.groupby('Category')['Size']  
        .mean()  
        .sort_values(ascending=False)  
)
```

Insight

Game-related categories generally have larger app sizes.

10. What are the top 5 categories with the most installs?

Approach

- Group by Category.
- Sum Installs.
- Sort descending.

Solution

```
top_categories_installs = (  
    df_cleaned.groupby('Category')['Installs']  
        .sum()  
        .sort_values(ascending=False)  
        .head(5)  
)
```

Insight

Games lead total installs followed by Communication and Social.

Intermediate Level Questions

1. Which categories have the highest average ratings?

Approach

- Group by Category.
- Calculate mean Rating.

Solution

```
top_categories = (  
    df_cleaned.groupby('Category')['Rating']  
        .mean()  
        .sort_values(ascending=False)  
        .head(10)  
)
```

Insight

Events, Education, and Art & Design rank highest.

2. How does app size relate to ratings?

Approach

- Scatter plot of Size vs Rating.

Solution

```
sns.scatterplot(  
    x='Size',  
    y='Rating',  
    data=df_cleaned  
)
```

Insight

No strong relationship exists.

3. Which apps receive the highest number of reviews?

Approach

- Sort by Reviews.

Solution

```
top_reviewed_apps = (  
    df_cleaned[['App', 'Reviews']]  
    .sort_values(  
        by='Reviews',  
        ascending=False  
    )  
    .head(10)  
)
```

Insight

Facebook and WhatsApp dominate review counts.

4. How do ratings vary across content ratings?

Approach

- Group by Content Rating.
- Calculate average Rating.

Solution

```
content_rating_avg = (  
    df_cleaned.groupby('Content Rating')['Rating']  
    .mean()  
)
```

Insight

Most content groups maintain similar ratings.

5. Analyze installs by category.

Approach

- Group by Category.
- Sum Installs.

Solution

```
category_installs = (  
    df_cleaned.groupby('Category')['Installs']  
    .sum()  
)
```

Insight

Games and Communication dominate.

6. Which categories generate the highest reviews?

Approach

- Group by Category.
- Sum Reviews.

Solution

```
category_reviews = (  
    df_cleaned.groupby('Category')['Reviews']  
    .sum()  
)
```

Insight

Social and Communication categories lead engagement.

7. Compare free and paid app ratings.

Approach

- Group by Type.
- Calculate mean Rating.

Solution

```
rating_by_type = (  
    df_cleaned.groupby('Type')['Rating']  
    .mean()  
)
```

Insight

Ratings are similar across free and paid apps.

8. Which categories have the highest install-to-review ratios?

Approach

- Aggregate installs and reviews.
- Calculate ratio.

Solution

```
ratio = (  
    category_installs /  
    category_reviews  
)
```

Insight

High ratios suggest broader reach per review.

9. Compare content ratings between free and paid apps.

Approach

- Cross-tabulate Type and Content Rating.

Solution

```
content_type_dist = pd.crosstab(  
    df_cleaned['Content Rating'],  
    df_cleaned['Type']  
)
```

Insight

Everyone dominates both free and paid apps.

10. Which genres have the highest install counts?

Approach

- Group by Genres.
- Sum installs.

Solution

```
genre_installs = (  
    df_cleaned.groupby('Genres')['Installs']  
    .sum()  
)
```

Insight

Action, Tools, and Communication dominate.

Advanced Level Questions

1. What are the top 10 apps with the highest ratings, and how do their reviews and installs compare?

Approach

- Sort by Rating and Reviews.

Solution

```
top_10 Rated = (  
    df_cleaned.sort_values(  
        by=['Rating', 'Reviews'],  
        ascending=[False, False]  
    )  
    [['App', 'Rating', 'Reviews', 'Installs']]  
    .head(10)  
)
```

Insight

High ratings do not always mean high popularity.

2. Analyze the trend of app updates over time.

Approach

- Extract year from Last Updated.
- Count apps per year.

Solution

```
updates_by_year = (  
    df_cleaned['Last Updated']  
    .dt.year  
    .value_counts()  
    .sort_index()  
)
```

Insight

Updates increased sharply after 2016.

3. How does average rating change with install count?

Approach

- Create install bins.
- Calculate mean rating per bin.

Solution

```
avg_rating_by_bin = (  
    df_cleaned.groupby('Install_Bin')['Rating']  
    .mean()  
)
```

Insight

Higher install ranges generally have higher ratings.

4. Perform sentiment analysis on app reviews.

Approach

- Requires review text dataset.
- Use NLP libraries such as TextBlob or VADER.

Solution

```
from textblob import TextBlob  
  
TextBlob(review_text).sentiment
```

Insight

Not applicable because review text is unavailable in the dataset.

5. What is the relationship between app genre and user ratings?

Approach

- Group by Genres.
- Calculate mean and median ratings.

Solution

```
genre_rating = (  
    df_cleaned.groupby('Genres')['Rating']  
    .agg(['mean', 'median', 'count'])  
    .sort_values(  
        by='mean',  
        ascending=False  
    )  
)
```

Insight

Creative, educational, and puzzle-related genres achieve the highest ratings.

Final Project Conclusion

The analysis reveals that:

- Free apps dominate the Play Store.
- Games generate the highest installs.
- Social and Communication apps drive the most engagement.
- Most apps maintain ratings above 4.0.
- App updates increased dramatically over time.
- High ratings do not necessarily indicate high popularity.
- Educational and creative genres achieve the best user satisfaction.
- Successful apps combine quality, engagement, frequent updates, and broad audience appeal.